



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III semester B.E. ECE B

Course Code & Title: EC8391 & Control Systems Engineering

Name of the Faculty member (s): P.Venkatesh

Innovative Practice Description

- **Unit / Topic:** IV / Nyquist Stability Criterion
- **Course Outcome:** CO4 : The students will be able to determine the stability of control systems using various methods
- **Topic Learning Outcome:** TLO 20: To determine the stability of the given system using Nyquist Stability Criterion
- **Activity Chosen:** Jigsaw
- **Justification:**

The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal each person depends on all the others.

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

The lessons are divided into subcategories. Students are divided into groups of four or five students. Then each small group would be created with one student receiving one Subcategory of the lesson. For this method, each small group gets the same set of Subcategories. Once individuals have researched their own subcategory, they will meet with individuals from the other small groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.

- **CO – PO / PSO mapping:**

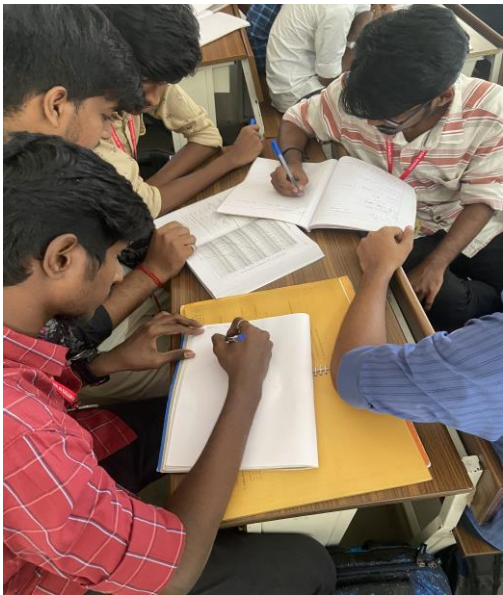
CO	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PSO1
CO4	3	3	2	2	1	1	1	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PSO1
	3	3	2	2	1	1	1	2
Justification for correlation	Determining the stability of a given system requires strong knowledge in advanced mathematical	The students will analyze the complex engineering problems and arrive	This PO is mapped to level 3 because students will refine	As the students use MATLAB to plot the root locus	During preparation of assignment and report/feedback on collaborative	The students will function effectively as a team to be	Presentation of the results and conclusions of the activity carried out, by the students,	This PSO has been mapped with level 2 as the methods

techniques like construction of Routh tables and communication engineering concepts and hence PO1 is mapped with level 3	the conclusion about stability of the system using mathematical principles.	the conceptual design of root locus into a detailed design using graphs with the same constraints	graph on giving the transfer function, PO5 has been mapped	activities, students should not indulge themselves in any unethical activities and hence this PO has been mapped	involved in group activities and also as an individual and hence this PO is mapped	require good communication and presentation skills and so PO 10 is mapped	available in the unit will help the students to solve existing stability oriented problems in control engineering
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• **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students feel that it was interesting. They learn how to communicate with team members and work together.

- ❖ **Benefit of the practice:**

Every student got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ **Challenges faced in implementation:**

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ https://www.educationworld.com/a_curr/strategy/strategy036.shtml
<https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>

- **Unit / Topic:** Unit V / Concepts of Controllability and Observability

- **Course Outcome:** CO 5

- **Topic Learning Outcome:** TLO 24

- **Activity Chosen:** Flipped Classroom

- **Justification:**

It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning. Also the practice was chosen because the topic is simpler and interesting for the students to refer and learn by themselves

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**

Specific topic was given to the students learn on their own. Resources like reference book, videos were given to the students. Apart from this they can refer any other relevant source for preparation. Students are asked to prepare more in depth than before. Students are separated into groups where students are given a specific topic to prepare and present. Get the class back together to share the individual group's work with everyone.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PSO1
CO5	3	2	2	1	1	1	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO8	PO9	PO10	PSO1
	3	2	2	1	1	1	2
Justification for correlation	Solve problems in topics like controllability require strong knowledge in mathematical topics like matrices and Numerical methods and hence PO1 has been mapped strongly to level 3	The students will analyze the engineering problems like controllability and observability of a given system and perform computations to solve the given mathematical model and so this PO is mapped moderately with level 2	PO3 is moderately mapped with level 2 because students will build and transfer models to determine the equivalence between state equations and transfer functions	Identify tenets of the professional code of ethics while doing activity based learning. So, it is mapped at level 2	It helps the students to present the topic of discussion as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 2	The students will present the results and conclusions of the activity carried out in front of the class which require good communication and presentation skills and so PO 10 is mapped	The students will solve existing/novel problems in control engineering.

• **Images / Screenshot of the practice:**





- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***

- Students feel that they have improved self-learning.
 - They learn how to communicate with team members and work together.
 - They were completely involved

- ***Benefit of the practice:***

- Every student got equal opportunity to come forward to take part in this activity.
 - They felt easy in solving level 3 questions in Assignments and Internal Assessment test
 - The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ***Challenges faced in implementation:***

- Few students hesitated to come forward to present the results and conclusions of the discussion
 - The main challenge faced is that few students not exposed to flipped class room.

References:

- ❖ <https://www.teachthought.com/learning/a-flipped-classroom/>
- ❖ Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021- 2022 (Odd Semester)

Degree, Semester & Branch : III Semester B.E. ECE B

Course Code & Title : CS8391 Control Systems Engineering (Unit I)

Name of the Faculty member : P.Venkatesh

Name of the topic	Translational system modelling
Name of the event	Finger signal
	

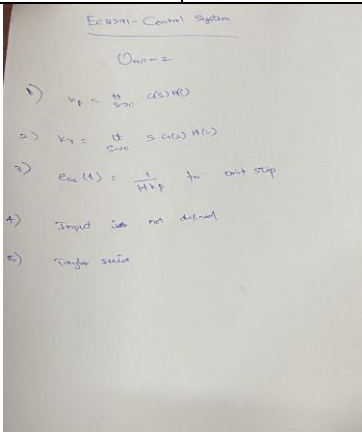
Name of the topic	Basic structure, feedback theory
Name of the event	Class poll
	

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Course Code & Title : CS8391 Control Systems Engineering (Unit II)

Name of the Faculty member : P.Venkatesh

Name of the topic	Steady error constant – static error & dynamic error
Name of the event	Quiz
	

Name of the topic	Type number, order, PID control
Name of the event	Open book test (library)
	

Signature of the Faculty member

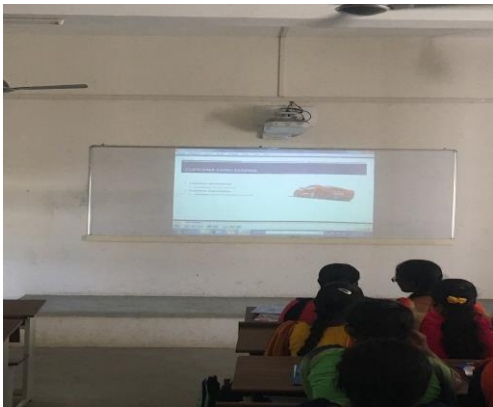
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Course Code & Title : CS8391 Control Systems Engineering (Unit III)

Name of the Faculty member : P.Venkatesh

Name of the topic	Frequency response of standard second order system-Bode Plot
Name of the event	Audio video tool
	

Name of the topic	Frequency response of standard second order system-Polar Plot
Name of the event	Buzz Session
	

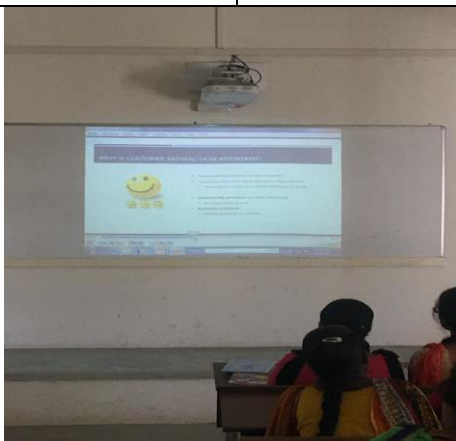
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Degree, Semester & Branch : III Semester B.E. ECE B

Course Code & Title : CS8391 Control Systems Engineering (Unit IV)

Name of the Faculty member : P.Venkatesh

Name of the topic	Introduction to Routh stability criterion
Name of the event	Zero minute speech
	

Name of the topic	Relative stability, Root locus concept
Name of the event	Audio video tool
	

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Degree, Semester & Branch : III Semester B.E. ECE B

Course Code & Title : CS8391 Control Systems Engineering (Unit V)

Name of the Faculty member : P.Venkatesh

Name of the topic	Stability of linear systems, Equivalence between transfer function and state variable representations
Name of the event	Open book test (Library)
	

Name of the topic	Concepts of Controllability and Observability
Name of the event	Exit slips
